

intervention strength, since it multiplies the steering vector’s contribution to the residual stream.² We perform hyperparameter tuning to select c and also the injection layer l . As expected from past work (Subramani et al. 2022; Mini et al. 2023), intervening at the middle layers is most effective.

Algorithm 1 and Figure 1 depict the resulting ActAdd method. In the appendix, Figure 6 illustrates a figurative example of steering a model with ActAdd if that model had one-dimensional residual streams (rather than e.g. GPT-2-XL’s 1600 dimensions). A runnable notebook can be found at tinyurl.com/actadd.

We test whether 1) steering vectors are effective at eliciting the desired behavioral shift, and 2) whether they preserve the general capabilities of the model. We run perplexity-based experiments on GPT-2-XL (1.5B parameters, Radford et al. 2019). We then run toxicity and sentiment experiments on OPT (Zhang et al. 2022b) and LLaMA-3 (Meta 2024).

4 RESULTS: ACTIVATION ADDITION WORKS

A summary of all experiments can be found in Table 5.³

4.1 ACTADD INTUITIVELY MODIFIES NEXT-TOKEN PROBABILITIES

We consider the OpenWebText corpus (Peterson et al. 2018). Our running example is the “wedding” topic vector produced by setting $p_+ = \text{weddings}$, $p_- = \text{''}$, $l = 16$, $c = 1$.

4.1.1 ACTADD REDUCES PERPLEXITY ON A TARGET TOPIC

For each document $d_i \in D$ in OpenWebText (Peterson et al. 2018), we first calculate the frequency of wedding-related words.⁴ If a document contains one of these words, the document is considered wedding-related. We randomly sample 300k documents, half of which are wedding-related.

We split the documents into sentences and measure GPT-2-XL’s perplexity on both the wedding-related and wedding-unrelated sentences. If the model is effectively steered to generate wedding-related text, it should assign that text higher probability (and thus achieve lower perplexity). For more details, see Appendix C.1.

Figure 2 shows the ActAdd perplexity relative to the unmodified model. In sentences where the injected topic (weddings) is more relevant, ActAdd’s perplexity is lower and predictive performance increases.

Figure 2: The perplexity ratio compares the relative predictive performance of ActAdd and an unmodified model. Lower is better. Adding the wedding steering vector improves performance on wedding-related text while preserving performance on unrelated text.



4.1.2 ACTADD’S IMPACT ON TOKEN PROBABILITIES

To test if the intervention is affecting relevant tokens or reducing perplexity in some spurious way, we observe the shift in the distribution of token log probabilities. We do this by randomly sampling 500 documents from the above OpenWebText sample and recording the log-probabilities assigned by the baseline and steered models. This results in a dataset of about 500k tokens, of which 29k are unique. We then group by token, filter for tokens with >20 instances in the dataset, and calculate the mean perplexity difference between the ActAdd and baseline models. By displaying these as a Q-Q plot (Gnanadesikan & Wilk 1968), we can inspect outlier shifts in token probability.

²It’s typical for the intervention strength c to have a magnitude less than 15.

³Code repository for our experiments: <https://zenodo.org/records/13879423>.

⁴wedding, weddings, wed, marry, married, marriage, bride, groom, and honeymoon.

Appendix Figure 8 shows the resulting mean log-probability difference distribution. We see that is approximately normal for the bulk of the tokens but with clearly heavy tails. The positive tail is significantly heavier than the negative tail, suggesting that one set of tokens are reliably increased in probability, with a smaller set of tokens reliably decreased to a lesser extent. Outlier tokens can be found in Appendix Table 11. *The probabilities most increased on average are primarily wedding-related.* The bottom tokens share no obvious theme and show a significantly lower absolute change in probability.

4.1.3 ACTADD STEERS THE MODEL TO DISCUSS WEDDINGS

At what layer are steering vectors most effective? Sweeping over GPT-2-XL injection layers for the wedding vector, we measure the average count of wedding-related words given a steering vector injected at each layer.

The intervention is already effective at the very first layer, rises in effectiveness until layer 6, and then declines. For the optimal injection site, we see >90% success in topic steering (compared to a ~2% baseline). Figure 3 shows the results of the layer sweep.

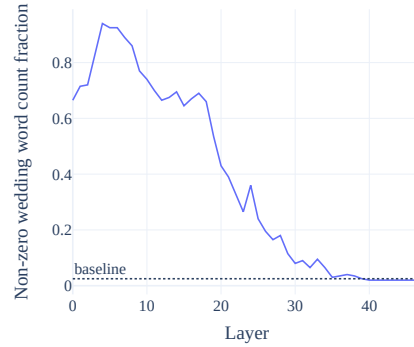


Figure 3: P(steered completion contains wedding-related words) as a function of injection layer.

4.2 ACTADD CAN CONTROL WHAT THE MODEL TALKS ABOUT

Method. Steering vectors can elicit generations on a range of topics – not just weddings. Starting from a generic prompt, we use GPT-3.5 to score whether the generations are about a target topic. Specifically, we generate 100 completions for the unmodified model and 100 for each target single-token ActAdd intervention (each token is about a different topic). Compared to the baseline generations, we record how much *more* frequently the steered model discusses the target topic.

Results. Figure 4 records a large boost in relevance (5-20%) on all topics at injection coefficient $c = 2$ (with the exception of “art”).

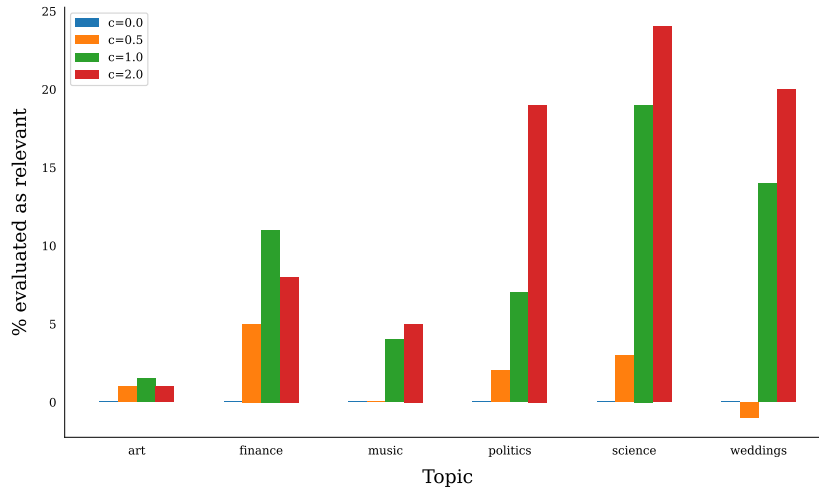


Figure 4: GPT-3.5-scored relevance of ActAdd completions on a range of generic topics.

4.3 ACTADD CAN REDUCE TOXICITY

Method. We benchmark toxicity reduction by generating steered continuations from RealToxicityPrompts (Gehman et al., 2020). Following Pei et al. 2023 we use a random subset $n = 1,000$. We repeat this sampling 5 times to obtain p -values (t -test against SOTA), bolding rows which are better with $p < 0.05$. For each continuation, we use the Perspective API to score toxicity.

Results. To establish a common scale, we reused the baselines and PREADD results from Pei et al. 2023, adding Air-Decoding Zhong et al. 2023 and FUDGE Yang & Klein 2021. This yields 6 baselines to compare ActAdd against. (We also considered Gu et al. 2022 (which reported 0.043 toxicity), but we could not reproduce the results; also, their disfluency (54.6) is too high for practical use.) We compare to ActAdd using OPT (Zhang et al. 2022b) and LLaMA-3 (Meta 2024).⁵

As shown in Table 3, ActAdd-OPT has 8% lower toxicity than the second-best, PREADD-D-OPT, and ActAdd-LLaMA-3 gives a 5% drop over LLaMA-3 with a very small fluency penalty.

Table 3: Results on RealToxicityPrompts (random $n=1000$). The OPT used is 6.7B parameters, LLaMA-3-8B. **Bold** is $p < 0.05$ against second-best. **Gray** text denotes numbers reported by Pei et al. 2023 (PREADD), Yang & Klein 2021 (FUDGE), or Zhong et al. 2023 (Air-Decoding). More recent models are less toxic by default. However, ActAdd-OPT is the least toxic of the OPT interventions and even outperforms an unsteered LLaMA-3.

Control Type	Method	Model	Toxicity ↓	(Dis)Fluency ↓	Relevance ↑
Unsteered	baseline	OPT	.134	8.9	.369
Prompting	baseline	OPT	.200	54.3	.294
Steering vector	ActAdd	OPT	.112	13.8	.329
Controlled gen.	FUDGE	GPT-2-M	.128	22.1	.329
Contrast. decoding	PREADD-S	OPT	.134	51.7	.290
Contrast. decoding	PREADD-D	OPT	.122	56.6	.326
Gradient-guided gen.	Air-Decoding	GPT-2-L	.185	48.3	-
Unsteered	baseline	LLaMA3	.114	6.3	.391
Steering vector	ActAdd	LLaMA3	.108	6.7	.365

4.4 ACTADD CAN CONTROL SENTIMENT

Method. To evaluate sentiment, we use the Stanford IMDB dataset (Maas et al., 2011). Our goal is for the model to continue each review but with the opposite sentiment. We compute the proportion of generated outputs with the desired sentiment, as classified by a model finetuned on sentiment data, SiBERT (Hartmann et al. 2023). For quality controls, we follow the conventional use of conditional perplexity to mark (dis)fluency, obtained using GPT-3 *davinci-002* logprobs. We use cosine similarity between the prompt and continuation sentence embeddings to gauge the relevance of text in $[0, 1]$. We evaluate sentiment changes from positive to negative and vice versa on a random subset of $n = 1,000$ and repeat to obtain p -values.

Results. Table 4 shows that our method is competitive on a conventional measure of sentiment control (Maas et al. 2011). We obtain state of the art success at steering from negative to positive sentiment. While. The only method which outperforms ActAdd in the positive to negative direction incurs a large penalty to fluency (68.4 vs 24.2, when matching methods on the same pretrained model) and relevance.

4.5 ACTADD PRESERVES THE MODEL’S GENERAL KNOWLEDGE

Method. We use ConceptNet from the LAMA benchmark, a general knowledge dataset (Petroni et al. 2019, $n = 29,774$ sentences, see Appendix Table 10). The model is given a prompt and then has to predict a factual completion. The task is intended for both causal and masked models, so some examples are difficult for causal-attention models (like GPT-2) due to the extremely limited context.

⁵We do not compare against finetuning because we wish to consider lighter-weight interventions which require minimal gradient updates.

Table 4: Results on IMDB sentiment. “Steering” denotes the probability of changing sentiment classification (called “success” in the baselines’ papers). **Bold** results represent $p < 0.05$ compared to the second-best. **Gray text** denotes numbers reported by Pei et al. 2023. *Underline* denotes best steered result. Fluency is worse under all steering methods; 1.5x to 3x worse for ActAdd, 7x worse for PREADD.

Method	positive to negative			negative to positive		
	Steering $\uparrow$	Disfluency $\downarrow$	Relevance $\uparrow$	Steer. $\uparrow$	Disflu. $\downarrow$	Rel. $\uparrow$
ActAdd-OPT	0.432	24.2	0.387	0.564	20.95	0.363
ActAdd-LLaMA3	0.268	8.6	0.354	0.669	15.2	0.275
OPT-Baseline	0.175	8.95	0.430	0.445	9.38	0.423
LLaMA3-Baseline	0.138	5.8	0.437	0.417	6.09	0.426
OPT-Prompt	0.307	53.5	0.298	0.365	50.9	0.287
FUDGE	0.532	25.1	0.311	0.551	22.7	0.320
PREADD-S-OPT	<u>0.631</u>	68.4	0.253	0.624	67.1	0.258

For each sentence, we run the model on its prompt with and without the wedding activation addition. $P@K$ is the probability that the expected label is among the model’s top- K predicted tokens, conditioned on the prompt. We score the baseline and modified models by calculating mean $P@K$ values for a range of K . Finally we plot these for both modified and unmodified models over a range of K values.

Results. Figure 5 shows that on the ConceptNet benchmark of factual questions, our method has a negligible impact on off-target answer probabilities.

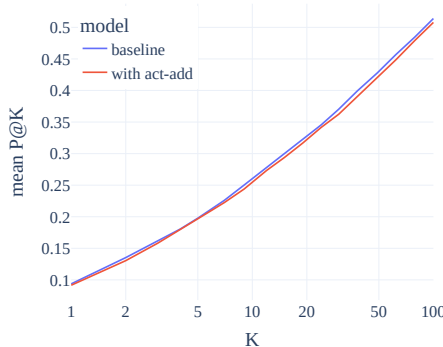


Figure 5: Testing side effects of ActAdd with the ConceptNet benchmark (Petroni et al. 2019). ‘ $P@K$ ’ is the probability of the correct answer being in the model’s top K answers. Our method has a negligible impact on off-target probabilities across a range of top- K values.

5 DISCUSSION

Algebraic combination of forward passes ActAdd can be viewed as composition of separate forward passes. For example, we compose $\mathbf{h}_+$, $\mathbf{h}_-$ and $\mathbf{h}^*$ to produce steered output. We were surprised that forward passes can “compose” in this way, despite the model not being trained to allow this operation. The composability of forward passes is itself evidence for compositional representations (Olah 2023), independent of the evidence from task-composition arithmetic on weights (Ilharco et al. 2023).

Limitations To steer the model using an ActAdd vector, the user supplies the injection coefficient c and the intervention layer l . So far we have had success with fixing the sequence alignment $a = 1$. Overall, these free hyperparameters make ActAdd less user-friendly than simple prompt engineering. Thankfully, the user does not have to perform a fresh hyperparameter sweep for each use case; in